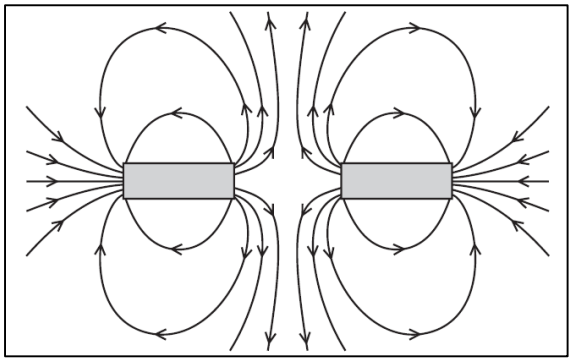
Magnetism

1. Law of magnetism:

Two same pole will.., while opposite pole will…

2. **N** part of compass will always show direction of (geographical south pole/ geographical north pole) because it is attracted by (magnetic south pole/magnetic north pole).



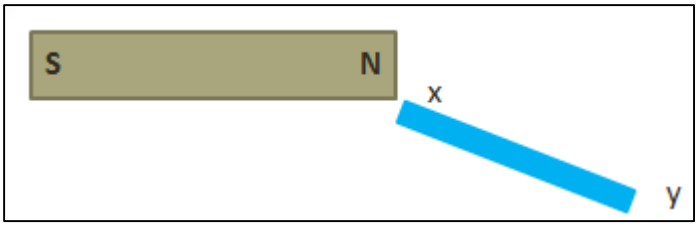
3. What is showed by arrow in this picture?
4. What happened when iron material is placed in the area filled with arrow?
5. Which one is S and N pole from this picture?

6. It is known that there are 3 specimens (X, Y, and Z). But it is unknown which specimen is made of **wood**, **soft iron**, or **steel magnet**. To identify it, we test the end A and end B of the specimen with permanent magnet.

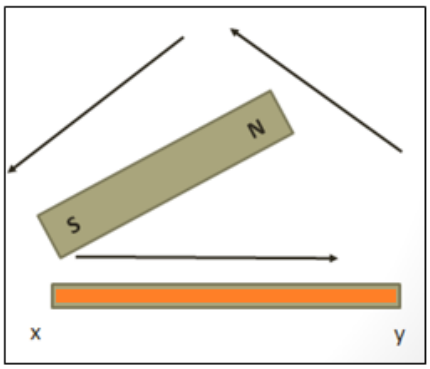
Below is the result of test. Fill the table with the correct material

Specimen	End A	End B	Material	Reason
X	No effect	No effect		
Y	Attracted	Repelled		
Z	Attracted	Attracted		

7. The blue object is a magnetic material, when it is placed near magnet, it will become an … magnets with x side become … pole, and Y side become … pole

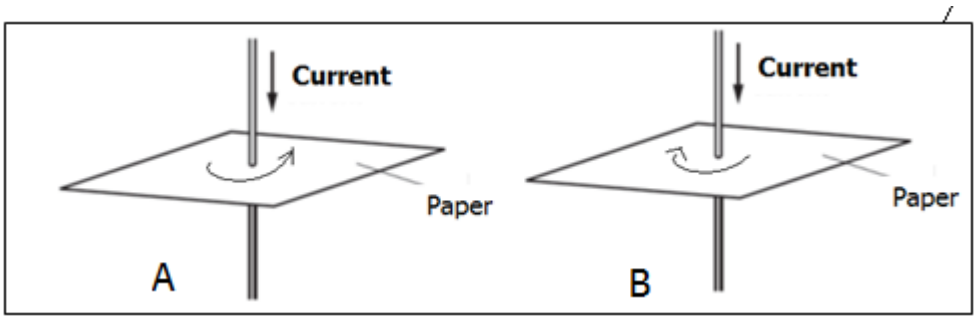


8. Orange rod is made of magnetic material. In this magnetization process, after several stroke **x** will become …pole, and **y** become … pole

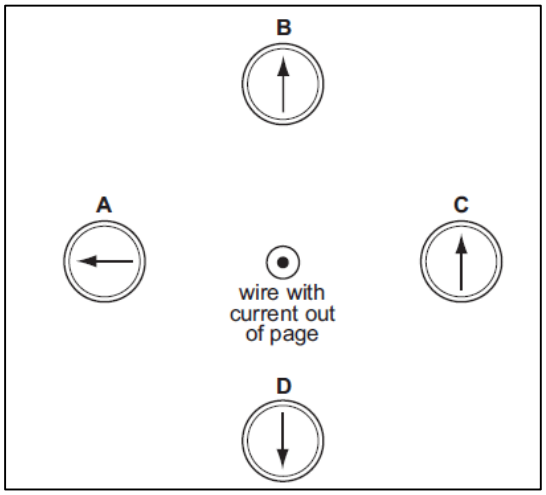


Electromagnet

1. Which is the correct drawing of magnetic field on the paper, because of electric current from the wire?



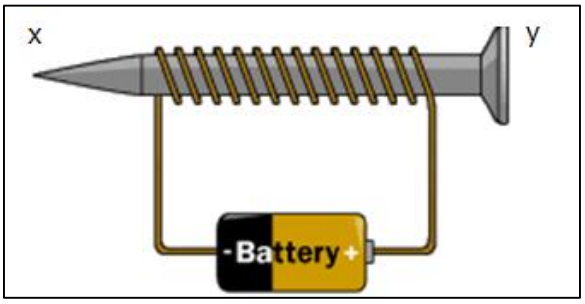
2. Which compass shows the correct direction of magnetic field generated by the wire



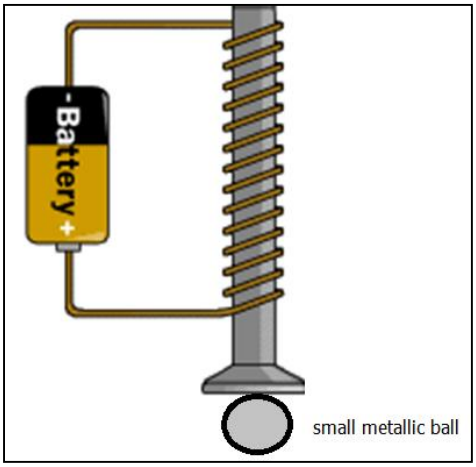
9. Electromagnet is magnet created by electricity. Why this kind of magnet is more useful?

10. Give two examples of electromagnetic used in daily life!

11. In this process part x will become … pole, and y become … pole



12. Why electromagnets use soft iron type of material instead of steel?

13. Suggest two ways to make electromagnet stronger!

14. The picture shows a small metallic ball is attracted to electromagnet (battery with iron nail). After several minutes, the battery is taken, but the small ball still attached. Explain why it happens!